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Getting ready for digital shift: the level of acceptance towards educational technology among faculty members in higher education institutions in Ethiopia

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Abstract

Introduction Understanding the extent to which faculty members' beliefs, attitudes, and acceptance of educational technologies for learning is crucial for enhancing the effectiveness of technology and ensuring its long-term viability. However, higher education institutions have made significant investments in educational technology for learning without fully comprehending faculty members' beliefs, attitudes, and level of acceptance of these technologies. This lack of understanding has hindered the effectiveness of these investments. Therefore, this study aims to examine faculty members' beliefs, attitudes and level of acceptance towards educational technology in higher education institutions in Ethiopia.

Methods A survey was conducted at five public higher education institutions that offer priority health training programs. The survey collected data using the Unified Theory of Acceptance and Use of Technology, focusing on four key determinants. Faculty members were asked to rate these determinants on a scale of 1 to 5. Each determinant was analyzed separately, examining the mean value and standard deviation. An overall mean score was calculated by combining all the determinants. Additionally, a logistic regression analysis was performed to determine how different demographic factors influenced faculty members' acceptance of technology for student learning.

Result A total of 330 faculty members participated in the study. The majority of respondents were male lecturers who held a second degree qualification. On average, the participants were 32.9 years old and had six years of teaching experience. Interestingly, nearly three fourth (72.6%) of faculty members hold positive beliefs and exhibit a high level of acceptance of educational technology. Furthermore, the likelihood of accepting and utilizing technologies for learning was found to be 2.3 times higher for faculty members working in teaching settings at research institutions.

Conclusion Faculty members have a favorable attitude towards educational technology, demonstrating a high level of acceptance. This positive belief holds significant implications and is crucial for enhancing the effectiveness of technology and ensuring its long-term viability.

Keywords Educational technology · E-learning · Ethiopia · Higher education institutions

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1 Introduction

Use of technologies for learning have gained momentum over the last decade within low-resource settings for the provision of education to health care providers due to its flexibility and accessibility [12,17,18,26]. Especially, due to the outbreak of COVID-19, there has been a significant increase in the use of technologies for learning in these countries [16]. The outbreak of COVID-19 especially changed the landscape of the educational systems and has made the learning institutions shift from the traditional face-to-face to the use of learning technologies [5,10,13,29,36].

The relationship between education and information technology in higher education institutions is intricate and transformative, shaping the learning landscape in profound ways. Here is an overview of key aspects the relationship between education and information technology: (1) Information technology has become an indispensable part of the teaching and learning process. By incorporating technologies like Learning Management Systems (LMS), online platforms, and multimedia resources, educators can greatly enhance the delivery of educational content [22]. (2) The advent of e-learning platforms and online education has revolutionized access to educational resources on a global scale. Virtual classrooms, webinars, and Massive Open Online Courses (MOOCs) utilize information technology to democratize education, making it accessible to learners worldwide [2]. (3) Information technology facilitates personalized learning experiences through the use of adaptive technologies. Learning analytics and artificial intelligence play a crucial role in customizing educational content to meet the unique needs of each student [28] 0.4) Information technology plays a crucial role in facilitating collaborative research endeavors among academics and students. Through the utilization of online research databases, collaboration platforms, and data analytics tools, higher education institutions are empowered to enhance their research capabilities. The advent of information technology has revolutionized the way academics and students engage in research. Online research databases provide a vast array of scholarly resources at the fingertips of researchers, enabling them to access a wealth of information from various disciplines. This accessibility not only saves time but also broadens the scope of research possibilities, fostering interdisciplinary collaborations and the exploration of new frontiers [7]

In conclusion, the relationship between education and information technology in higher education institutions is dynamic, influencing pedagogy, accessibility, research, and administration. As technology continues to evolve, its impact on education is likely to deepen, necessitating ongoing research and adaptation.

However, the effective utilization of technology for learning requires faculty members who embrace educational technology, and possess proficiency in using digital tools [8]. The acceptance and utilization of educational technology by faculty members is pivotal in today's rapidly evolving digital landscape. As technology continues to advance, it is increasingly imperative for educators to wholeheartedly adopt these tools and seamlessly integrate them into their teaching practices [27].

1.1 Theoretical perspective of technology acceptance

Numerous theories and models have been developed to explain and predict the effective utilization of information system technologies by educators. Among these, the Unified Theory of Acceptance and Use of Technology (UTAUT) stands out as a comprehensive synthesis of previous research on technology acceptance, devised by Venkatesh, Morris, Davis, and Davis.

The UTAUT serves as a valuable framework for understanding the factors that influence educators' acceptance and utilization of technology in their educational practices. UTAUT proposes four fundamental constructs that serve as the key factors influencing the acceptance and utilization of technology. These four core constructs are performance expectancy(PE), effort expectancy(EE), social influence(SI), and Facilitating Conditions(FC) [34].

PE refers to the extent to which an individual believes that using an information system will help him or her to attain benefits in job performance [22,30,34]. EE is defined as the degree of ease associated with the use of technology [30,34]. Based on the UTAUT, use of technology among educational users will depend on whether or not the technology is easy to use [9,22,34]. SI: Venkatesh, Morris, Davis & Davis defined SI as the degree to which a person perceives how important it is that "other people" believe he or she should use a technology [9,22,34]. FCs are defined as the extent to which users believe that the necessary infrastructures to support the use of technology in an organization exist. These may include resource and technology factors concerning compatibility issues that have an impact on usage [9,22,30]. FC also includes the necessary training for the technology users. FC is the organizational and the technical support for the users (Supplementary Figure 1).

Understanding the extent to which a faculty members(FMs) believes and attitudes on PE, EE, SI, and FC by Higher Education Institutions (HEIs) is crucial for the acceptance and usage of technologies for learning and can greatly benefit HEIs in several ways. Firstly, performance expectancy refers to the perceived usefulness and effectiveness of a technology in enhancing learning outcomes. By understanding PE, HEIs can identify which technologies are most likely to meet the needs and expectations of their students and faculty. This knowledge allows them to invest in the right tools and resources that align with their educational goals, leading to improved student engagement, satisfaction, and academic performance. Effort expectancy focuses on the ease of use and user-friendliness of a technology. HEIs need to assess EE to ensure that the chosen technologies are intuitive, accessible, and require minimal training or technical expertise. By considering EE, HEIs can select user-friendly platforms that reduce barriers to adoption, encourage active participation from both students and faculty [22].

Higher education institutions in Ethiopia have made significant investments in educational technology for learning. However, higher education institutions have made significant investments in educational technology for learning without fully comprehending faculty members' beliefs, attitudes, and level of acceptance and utilization of these technologies. This lack of understanding has hindered the effectiveness of these investments [3,14,31]. Previous studies [11,24,33] has highlighted the importance of comprehending the key factors that drive faculty members' acceptance of technology for learning. To enhance the acceptance and usage of educational technologies, it is crucial for institutions to identify and address the factors that influence their adoption. By doing so, they can ensure that these investments are utilized to their full potential, benefiting both students and educators. Therefore, this study aims to examine faculty members' beliefs, attitudes and level of acceptance towards educational technology in higher education institutions in Ethiopia, allowing for proactive intervention design.

2 Materials and methods

2.1 Study design

We employed a cross-sectional survey to assess the acceptance and utilization of technology in education. We considered the advantages of survey designs, such as their cost-effectiveness and efficient data collection process. In this survey, data was collected at a single point in time to draw generalizations from a sample of a larger population, enabling us to make inferences about the population as a whole.

2.2 Study setting and target populations

The study was conducted from December 15, 2022 to February 30, 2023 at public HEIs that offer the six priority health training programs, namely medicine, nursing, anesthesia, midwifery, laboratory, and pharmacy in Ethiopia. These programs have been identified by the Ministry of Health as priority health programs. The study focused on HEIs located in the three largest regions of the country, which collectively represent the majority of the country. These HEIs are currently classified into three distinct categories: research universities, universities of applied science, and comprehensive universities. Research universities have a primary mission of conducting cutting-edge research and delivering advanced post-graduate teaching. Universities classified as applied science primarily focus on applied science, particularly in undergraduate studies. Comprehensive universities, on the other hand, place a strong emphasis on providing a well-rounded general education [19]. The target population of the study was the faculty members(FMs) involved in the six priority health programs at these selected institutions. At the time of study, there were 16 public HEIs that offer these priority academic programs.

2.3 Sampling size and methods

We used a sample size of 30% (i.e. 30 are often considered sufficient for the Cental Limt Theory to hold [23] to select target institutions. As a result, we selected five institutions from a total of sixteen using a simple random sampling method. Since there is no universally available data on the proportion of e-learning acceptance among university teachers in Africa, we assumed a population proportion of 0.5 with a 95% confidence level and a margin of error of

5% to determine the sample size required to estimate the proportion of faculty members accepting e-learning at HIEs. The equation for calculating sample size n (with finite population correction) = [z^2 * p * (1−p) / e^2] / [1 + (z^2 * p * (1−p) / (e^2 * N))] Where:

n is the sample size,
z is the z-score associated with a level of confidence (1.96)
p is the sample proportion (50%)
e is the margin of error (0.05)
N is the population size (1377)

In order to ensure statistical accuracy, a minimum sample size of 301 individuals is required. Additionally, to account for a potential non-response rate of 10%, a total of at least 331 participants would be necessary. consequently, a random sample of 331 individuals was chosen from a total of 1377 individuals in five organizations. This was done by assigning unique identifiers to each person, generating random numbers, sorting them, and selecting the corresponding individuals.

2.4 Data collection methods and instrument

A survey was conducted using the UTAUT model to evaluate the acceptance and usage of technology across four key determinants that drive acceptance of educational technologies. The UTAUT model is a useful tool for assessing the likelihood of success for new technology introductions and helps in understanding the factors that drive acceptance, thereby enabling the design of proactive interventions [22]. The UTAUT model proposes four fundamental constructs that play a crucial role in influencing the acceptance and utilization of technology. Each construct is measured using a specific number of items: PE consists of 7 items, EE consists of 6 items, SI consists of 6 items, and FC consists of 6 items. All 25 items within the four constructs were assessed using a five-point Likert scale, which ranged from 1 (strongly disagree) to 5 (strongly agree) [34]. The UTAUT model was tested, validated and resulted in an R^2 value of 70%, indicating that the model explains 70% of the variation in user intentions to use information technology [6,35]. Data collection took place from December 15, 2022 to February 30, 2023, with trained and experienced data collectors using the Open Data Kit (ODK) collect open-source Android app. This app is freely available for survey-based data gathering purposes [21]**.**

2.5 Data analysis

The ODK data was imported into SPSS version 25 and checked for accuracy and completeness. Descriptive statistics (frequency and percentage) were used to describe and visualize the characteristics of the participants. The study analyzed each determinant factor (PE, EE, SI, and FC) that influences the acceptance and usage of educational technologies for learning. The mean value and standard deviation of each factor were examined separately, and an overall mean score was calculated by combining all the factors. Since the scale consisted of 25 items and the highest possible score for a participant was 125, the scores received by participants were categorized into three groups: low (scores between 0 and39), average (scores between 40 and 79), and high (scores between 80 and125) [22]. Furthermore, a logistic regression analysis was performed to assess the impact of different demographic factors on the likelihood of FMs expressing a high level of acceptance towards technologies for student learning.

3 Result

3.1 Sociodemographic characteristics

A total of 330 FMs participated in this study (response rate = 99.7%). The majority of the respondents were male, accounting for 279 (84.6%), and they were employed in comprehensive level of HEIs 139 (42.1%). Majority of the respondents holds Second Degree qualification 246 (74.6%) and Lecturer position 193 (58.5%). Respondents had a median teaching experience of 6 years and were 32.9 (SD 5.3) years old on average (Table 1).

Table 1 Background characteristics of study participants (n = 330)

Variables	Response	Frequency	Percentage
Level/Type of University worked in	Research University	100	30.3
	Applied University	91	27.6
	Comprehensive	139	42.1
Academic program	Medicine	103	31.1
	Nursing	75	22.8
	Midwifery	37	11.3
	Anesthesia	33	10.0
	Medical Lab	40	12.2
	Pharmacy	42	12.6
Sex	Male	279	84.6
	Female	51	15.4
Academic qualification(Highest)	First degree (BSc/MD)	24	7.2
	Second degree (MSc/MPH)	246	74.6
	Specialty & sub-specialty certificate (MD plus)	47	14.3
	Third Degree (PhD)	13	3.9
Academic rank	Lecturer	193	58.5
	Assistant Professor	127	38.5
	Associate Professor	9	2.6
	Professor	1	0.4
Mean age in years (SD)	32.9 (5.3)		
Median teaching experience in HEIs in years (IQR)	6 (5)		

Table 2 The mean scores achieved for each factor in the UTAUT model and the overall mean score

UTAUT factors variable categories	Number of items	Mean score of items	SD
Performance expectancy	7	4.20	0.56
Effort expectancy	6	3.87	0.65
Social influence	6	3.6	0.71
Facilitating conditions	6	3.36	0.76
Over all mean score	25	3.75	0.59

The majority of the scale items yielded scores ranging from "agree" to "strongly agree," with occasional instances of a neutral response (Supplementary information Table 1). Notably, participants displayed a generally positive belief and attitude towards acceptance and usage of educational technologies for learning, as indicated by the mean scores for each determinant factor: PE ($\bar{x} = 4.20$), EE ($\bar{x} = 3.87$), SI ($\bar{x} = 3.6$), and FC ($\bar{x} = 3.36$). Moreover, the overall technology acceptance mean score ($\bar{x} = 3.75$) exceeded the "average" level on the scale, suggesting an overall positive belief and attitude towards technology acceptance and usage among faculty members (Table 2).

3.2 Over all technology acceptance level

The scores received by participants were categorized as low (scores between 0 and 39), average (scores between 40 and 79), and high (scores between 80 and 125). The objective was to determine the overall acceptance level of technology among FMs [22]. The findings unveiled an impressive trend, with almost three-quarters of FMs scoring high level in technology acceptance. Remarkably, none of the FMs scored low in terms of their acceptance and utilization of technology for learning purposes (Table 3).

Table 3 FMs demonstrate varying levels of technology acceptance—categorized as low, average, and high scores

Score category	Number (%) of FMs who scored:	%	Low level of acceptance and usage of technologies for learning
0 -39	0 (0.00)	0.00	Low
40–79	90 (27.3)	27.3	Average
80–125	240 (72.7)	72.7	High
Total	330 (100)	100	

A logistic regression analysis was conducted to evaluate the influence of various demographic factors on the probability of FMs reporting a high level of acceptance towards technologies for student learning. The goodness-of-fit test was conducted by comparing the *p*-value of the Hosmer–Lemeshow test statistic with a predetermined significance level α. Based on the Hosmer–Lemeshow test statistic and its *p*-value of 0.73, there is no significant evidence of lack of fit in this model. Therefore, it was reasonable to proceed with further analysis of this model. Furthermore, the chi-square value for the Hosmer–Lemeshow Test is 5.256, with a significance level of 0.73. This value exceeds 0.05, indicating support for the model fit.

Only two of the independent variables, the age of FMs and the teaching setting, had a statistically significant effect ($p < 0.05$) on the level of technology acceptance and usage for learning. The strongest predictor of reporting a high level of technology acceptance for learning was the teaching setting, with an odds ratio of Exp(B) = 0.433. This shows that when Facility Managers (FMs) work in teaching settings of applied or comprehensive institutions, their likelihood of accepting technology decreases by 56.7%, compared to FMs who work in teaching settings of research institutions, keeping that all other factors remain constant. This effect is statistically significant ($p < 0.001$). In other words, the likelihood of accepting and using technologies for learning is 2.3 times higher for FMs working in teaching settings at research institutions.

Additionally, the odds ratio for FMs age is 1.073, implies that for every additional year of age, the chances of accepting and using technologies for learning increase by a factor of 1.073. This effect is statistically significant, with a p-value of 0.03 and a 95% confidence interval of [1.005, 1.145] (Table 4).

4 Discussion

The acceptance and use of educational technology by faculty members is crucial in today's rapidly evolving digital landscape. As technology continues to advance, it has become increasingly important for educators to embrace these tools and integrate them into their teaching practices [27]. Understanding the extent to which a FMs attitude and believes on key determinant factors that drive acceptance of technologies for learning by HEIs is crucial for enhancing the effectiveness of technology and ensuring its long-term viability. This study aims to examine the extent to which a faculty member believes on key determinant factors that drive acceptance and usage of educational technologies for learning. The findings of this study indicate

Table 4 Logistic regression predicting likelihood of reporting a high level of acceptance and usage of technologies for learning

Variables	B	S.E	Wald	df	Sig	Exp(B)	95% C.I. for EXP(B)	
							Lower	Upper
Sex (male*)	0.335	0.319	1.102	1	0.294	1.398	0.748	2.615
Age in years	0.071	0.033	4.515	1	0.034**	1.073	1.005	1.145
Academic rank (lecture*)	−0.136	0.258	0.280	1	0.597	0.872	0.526	1.447
Teaching setting (Research HEIs*)	−0.837	0.262	10.230	1	0.001**	0.433	0.259	0.723
Educational qualification (First degree*)	0.534	0.404	1.744	1	0.187	1.706	0.772	3.767
Year of experience	0.023	0.030	0.599	1	0.439	1.023	0.965	1.084
Constant	1.616	0.998	2.620	1	0.106	0.199		

*Reference category

**Statistically significant at 0.05 level of significance

that nearly three out of four of faculty members hold positive beliefs and exhibit a high level of acceptance of educational technology. This high level of acceptance and positive belief can have various implications for the overall educational experience and outcomes. When faculty members have positive beliefs about educational technologies, they are more likely to use and incorporate these tools into their teaching practices. Research has demonstrated that faculty members who view technology as beneficial and believe it improves student learning are more inclined to integrate it into their courses [25,34].

On the other hand, the positive beliefs and high level of acceptance of educational technologies among faculty members can have a significant impact on student engagement and motivation. By incorporating technology tools and platforms into their teaching, instructors can offer students more interactive, multimedia-rich, and personalized learning experiences. This, in turn, can enhance student engagement and motivation, encouraging them to actively participate in the learning process [20]. Furthermore, the use of technology can facilitate deep learning, critical thinking, and problem-solving skills among students, ultimately leading to improved learning outcomes [4]. Faculty members who have positive perceptions of technology often report increased engagement, interactivity, and collaboration among students [1]. This increased engagement can lead to improved motivation, participation, and ultimately better academic performance.

In addition, faculty members who have a positive view and high level of educational technologies acceptance are more inclined to explore and try out various teaching methods that are supported by technology. They are open to participating in professional development opportunities and gaining the skills needed to effectively incorporate technology into their teaching. This can lead to the creation of innovative and student-focused instructional approaches [15]

Teaching setting and the individual characteristics of faculty members play a crucial role in their acceptance and utilization of educational technologies [1, 37]. This study found a significant association between the high level of acceptance of technologies and the specific type of institution in which FMs are worked in. For instance, the likelihood of accepting and using technologies for learning is 2.3 times higher for FMs working in teaching settings at research institutions. These research institutions, with their primary focus on conducting groundbreaking research and providing advanced post-graduate teaching, foster an environment that encourages the integration of cutting-edge technologies. This could be attributed to the presence of suitable technological infrastructure, resources, support, training, and institutional policies and culture in these well-established institutions. Studies have shown that the availability of appropriate technological infrastructure and resources, institutional policies and culture [37], and adequate support and training [32] are crucial for faculty members to effectively utilize educational technologies.

5 Significance of the study

Studying the acceptance of educational technologies by faculty members is crucial for comprehending the advantages and consequences of integrating these technologies in higher education. By gaining insight into faculty acceptance, institutions can provide the necessary support and resources to promote technology-enhanced teaching practices. Gaining insight into individuals' attitudes, perceptions, and behaviors towards these technologies can greatly aid institutions and academic leaders in making well-informed decisions. Additionally, it can assist institutions in tailoring strategies to effectively adopt e-learning approaches.

6 Strength and limitation of the study

This study boasts a significant strength in its comprehensive national survey, encompassing a wide range of HEIs across various geographical locations. This extensive coverage enables the generalization of results to a larger population, enhancing the study's credibility and relevance. However, it is crucial to acknowledge a limitation inherent in survey-based research. The reliance on self-reported data introduces the potential for respondents' subjective interpretations or misunderstandings, which can undermine the reliability of the collected data.

7 Conclusion

It is evident that faculty members possess a favorable attitude towards educational technology, demonstrating a high level of acceptance. This high level of acceptance and positive belief can have various implications for the overall educational experience and outcomes. When faculty members have positive beliefs about educational technologies, they are more likely to use and incorporate these tools into their teaching practices. This positive belief and high level of

acceptance among faculty members can have a profound impact on student engagement and motivation. Furthermore, faculty members who possess a positive perspective and a strong acceptance of educational technologies are more likely to explore and experiment with different teaching methods that are supported by technology. This understanding is crucial for enhancing the effectiveness of technology and ensuring its long-term viability.

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Author contributions All authors contributed to the study's conception, design, and data collection. Data cleaning and analysis were performed by EM and RT. The first draft of the manuscript was written by EM and RT, and all authors commented on the first versions of the manuscript. All authors read and agree to be accountable for all aspects of the work, ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

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Data availability The dataset that supports the analysis and interpretation of findings from the current study is available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate The research protocol, with reference number JUIH/IRB/266, was approved by Jimma University's Institute of Health. The research was conducted in accordance with the guidelines set forth by the Institutional Review Board, as outlined in the ethics statement. Participants were provided with informed consent to participate in the study, and they willingly expressed their agreement to take part orally. This decision was made based on the understanding that the study presented no significant risks or consequences.

Competing interests The authors declare no competing interests.

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